

Question1

Use induction to prove that $6|(15n + 9n^2)$ for any $n \in \mathbb{Z}^*$

Question 2

Suppose that p is a prime number with $p \geq 5$.

1. Show that $p \equiv 1 \pmod{6}$ or $p \equiv 5 \pmod{6}$

2. Show that $p^2 + 2$ cannot be prime

Question 3

Suppose that $x, y, d \in \mathbb{Z}^+$. Show that if $d \neq 0$ and $dx|dy$, then $x|y$.

Suppose that $a, b, d \in \mathbb{Z}^+$, Show that if $d = \gcd(a, b)$, then $\gcd\left(\frac{a}{d}, \frac{b}{d}\right) = 1$

Suppose that $a, b, c, n \in \mathbb{Z}^+$, and that $ca = cb \pmod{n}$. Show that $a = b \pmod{n}$, where $d = \gcd(c, n)$

Question 4

Consider the set $S = \{A \in Z, 2 \in A\}$, and let $P(Z \setminus \{2\})$ denote the power set of $Z \setminus \{2\}$. Show that the map

$$f: P(Z \setminus \{2\}) \rightarrow S$$

Defined by $f(A) = A \cup \{2\}$ is a bijection.

Show that S is uncountable. Note: You may assume $\mathbb{Z} \setminus \{2\}$ is countable infinite without proof.